Pair-index relation reuse strengthens the generic baseline

2026-10-30. Verified finite stage result; **same-total-budget superiority remains unestablished**. This continues the existing Hecke/index-calculus goal and follows merged PR1317. The four original field/point/target fixtures are unchanged.

Measured information

The signed pair index required for the original target oracle is now built before solving base logs. When P_i+P_j=P_k+P_l, its collision gives a valid linear relation. Known negation/zero identities are removed after coefficient folding. Retaining these collisions uses exactly the original curve sums; coefficient bookkeeping and elimination are additional work, not free CPU time.

Subgroup r / Folded columns / Previous Hecke/scalar free logs / Pair-only rank / free logs / Hecke + pair rank / free logs / Marginal Hecke rank over pair-only

769 / 32 / 9 / 31 / 1 / 31 / 1 / 0

16,319 / 32 / 12 / 21 / 11 / 31 / 1 / 10

206,477 / 32 / 12 / 1 / 31 / 20 / 12 / 19

46,273, conductor30 / 159 / 45 / 158 / 1 / 158 / 1 / 0

Thus pair-index reuse lowers required generic log queries9→1,12→1, and45→1 on three fixed cells. It does not reduce the12 remaining queries on the sparsest cell. The earlier doubling/tripling-only generic comparison is a weaker control; its original measured values remain correct for that method, but must not be read as the best available generic preprocessing baseline. In particular the conductor30 pair-only system already has rank158, so Hecke supplies no further rank there under this stronger comparison.

Hecke still supplies10 and19 marginal constraints on the medium/large f6 cells, respectively. This is information beyond this complete pair-collision collector, not a lower bound against every same-budget generic algorithm. The extra cost of generating the Hecke relations has not been fully priced.

Costs, controls and target behavior

Index construction still uses2145 additions for each f6 cell and51040 for conductor30, exactly matching the original index. Its input point set and, on the large field, its point-set digest match the frozen arity-control index. Every unordered signed pair is enumerated, including duplicates, even after the retained relation basis reaches maximal possible rank.

The frozen sqrt(r)-sized BSGS table is unchanged across arms. Counts below cover generic base-log queries and each query's scalar replay, not geometric construction, matrix bookkeeping, or final all-label replay.

r / Previous Hecke/scalar query additions / Generic pair-only query additions / Hecke + pair query additions

769 / 219 / 33 / 33

16,319 / 858 / 897 / 94

206,477 / 3836 / 9902 / 3836

46,273 / 5941 / 73 / 73

Fewer generic queries do not automatically mean proportionally fewer additions: the generic medium-field arm uses11 queries but897 additions versus the previous12-query arm's858 because different free points require different search lengths. Full costs and qualified speed ratios remain unknown.

All three arms on all four cells recover identical base labels and preserve the original target statuses: three successes at the same attempt counts, and the large-field256-attempt failure. Changing label preparation leaves the base point set and target coverage unchanged. No additional target decomposition or fresh-target success is claimed.

Reproduction and verification

PAIR_REUSE_PROTOCOL.md was committed locally before enumeration. Publication of that commit was delayed by a concurrent merge into the shared PR branch; it was preserved and merged without force-pushing. These are exploratory fixed-fixture controls, not an independently public holdout. No earlier raw receipt was modified.

Run `run_pair_reuse.py --out-root UNIQUE_DIRECTORY` through the checkpoint gate. It retains all four attempts, including failures, with600-second cell and2400-second suite caps. runs/pair-reuse-01 contains the outputs, source/input hashes, group counters, collision-basis rows and explicit four-point witnesses. A separate short-Weierstrass implementation re-enumerates all57,475 pairs and independently checks the rank of all21,118 nonzero collision rows, not just the retained basis. It also replays all765 base-label checks across the three arms (255 points per arm).

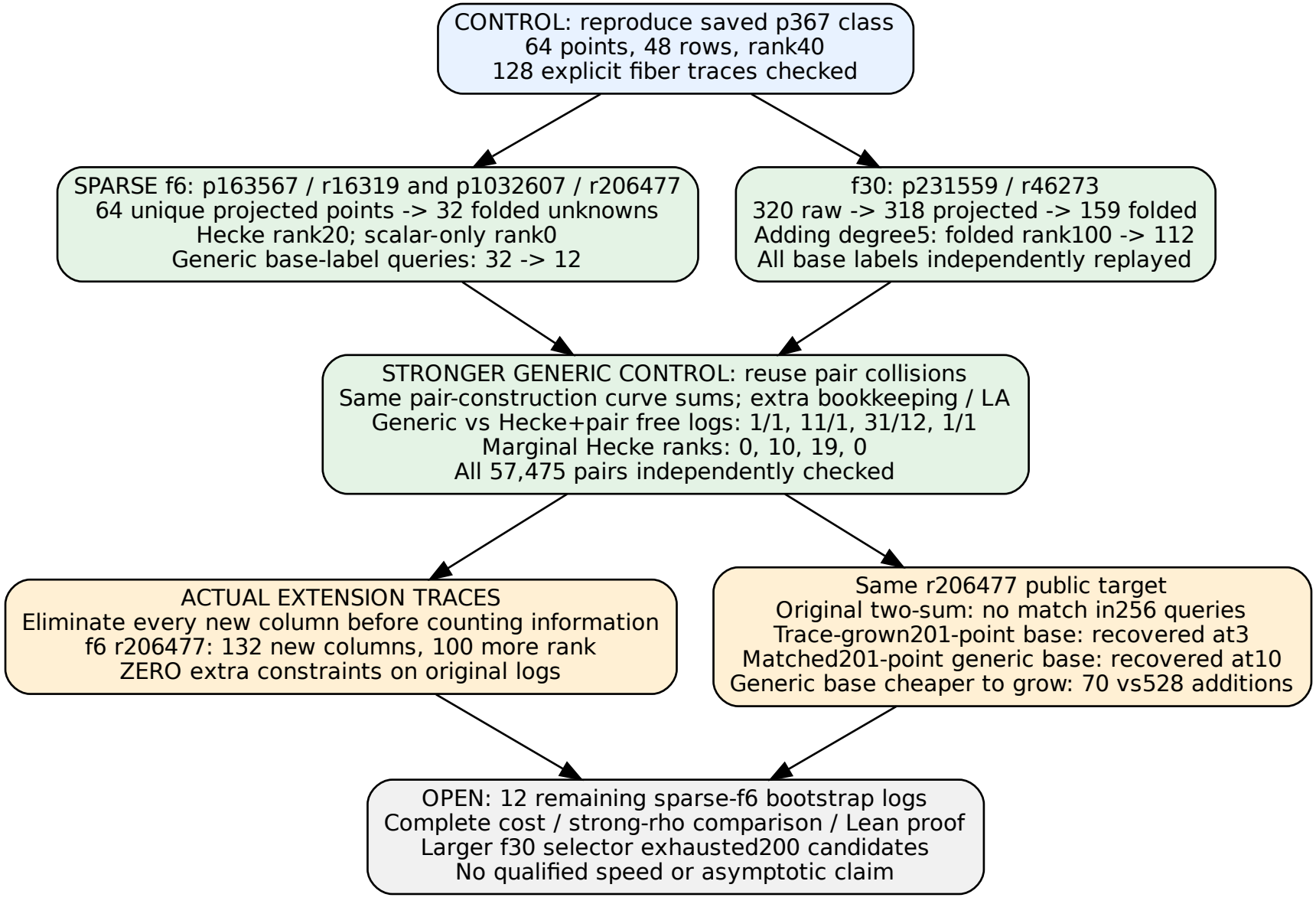
Two additional unit tests check dependent/independent coefficient rows, a hand-known five-torsion collision, zero identities, and rejection of a corrupted witness. The earlier four arithmetic tests remain required. Two cost-workload tests additionally check complete BSGS search under tiny table caps and fail-closed handling of an absent CUDA device. All eight pass; the prepared cost workload itself has not run on an accepted benchmark host. The geometric interpretation and Lean limitations remain as stated in PROOF_MAP.md; no new global theorem is claimed.

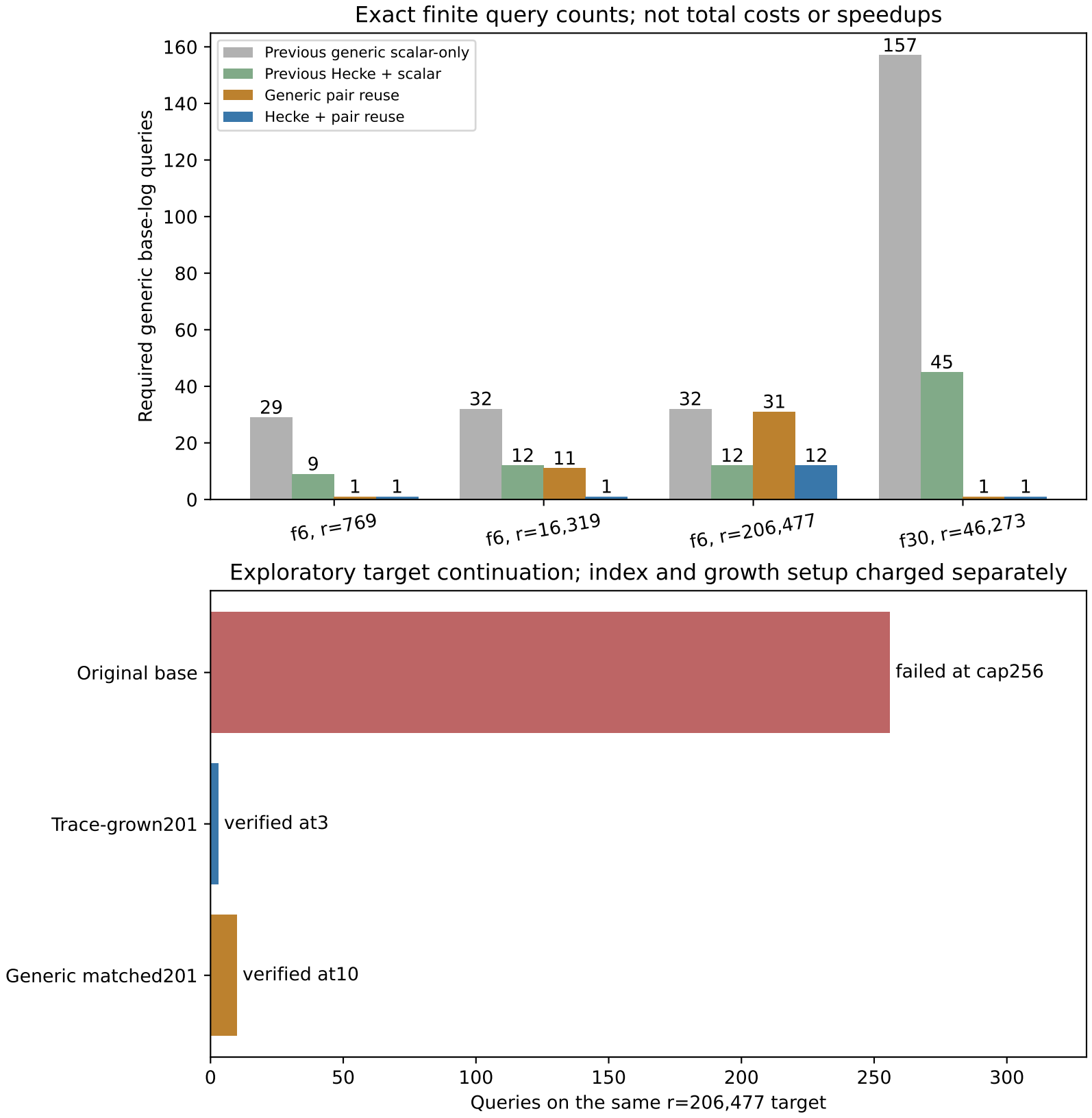
Remaining blocker for the external goal

The requested same-total-budget advantage is still not established. It needs a declared complete cost boundary that charges geometric function/fiber construction, coefficient tracking, LA, bootstrap and target work for both arms, along with stronger generic collectors and fresh single-target controls. Reusing a table gives a verified improvement in query preparation within this pipeline; it does not settle that comparison.

The selected environment has no NVIDIA device or configured RunPod/Modal/fleet compute binding; access evidence records presence checks without secret values. The repository's AGENTS.md states: "Index-calculus runs, benchmarks and measurements execute on a CUDA GPU host." Qualified cost/rho measurement is blocked on that execution capability; ordinary-host correctness diagnostics do not substitute for it. This is an access/measurement gap, not a proof that a Hecke advantage is impossible. The source problem record retains the goal.

COST_CELL_PROTOCOL.md and cost_cell.py make the next fixed-base discriminator concrete: charge geometry, pair bookkeeping, LA, a tuned shared BSGS table, recovery and label replay. The generic control uses the same table-memory cap. Full base generation and fresh target descent remain separate pipeline costs. The preflight attempt exited with status2 and `benchmark_executed=false` because no CUDA device was accessible; it ran no benchmark arithmetic. Execution and qualification of this workload remain blocked. An accepted harness must also supply isolation, calibration, alternating repetitions, A/A and positive controls before any speed ratio.


CONTROL: reproduce saved p367 class
64 points, 48 rows, rank40
128 explicit fiber traces checked
SPARSE f6: p163567 / r16319 and p1032607 / r206477
64 unique projected points -> 32 folded unknowns
Hecke rank20; scalar-only rank0
Generic base-label queries: 32 -> 12
f30: p231559 / r46273
320 raw -> 318 projected -> 159 folded
Adding degree5: folded rank100 -> 112
All base labels independently replayed
STRONGER GENERIC CONTROL: reuse pair collisions
Same pair-construction curve sums; extra bookkeeping / LA
Generic vs Hecke+pair free logs: 1/1, 11/1, 31/12, 1/1
Marginal Hecke ranks: 0, 10, 19, 0
All 57,475 pairs independently checked
ACTUAL EXTENSION TRACES
Eliminate every new column before counting information
f6 r206477: 132 new columns, 100 more rank
ZERO extra constraints on original logs
Same r206477 public target
Original two-sum: no match in256 queries
Trace-grown201-point base: recovered at3
Matched201-point generic base: recovered at10
Generic base cheaper to grow: 70 vs528 additions
OPEN: 12 remaining sparse-f6 bootstrap logs
Complete cost / strong-rho comparison / Lean proof
Larger f30 selector exhausted200 candidates
No qualified speed or asymptotic claim


Exact finite query counts; not total costs or speedups
Previous generic scalar-only
Previous Hecke + scalar
Generic pair reuse
Hecke + pair reuse
Required generic base-log queries
0
20
40
60
80
100
120
140
160
29
9
1
1
32
12
11
1
32
12
31
12
157
45
1
1
f6, r=769
f6, r=16,319
f6, r=206,477
f30, r=46,273
Exploratory target continuation; index and growth setup charged separately
Original base
failed at cap256
Trace-grown201
verified at3
Generic matched201
verified at10
0
50
100
150
200
250
300
Queries on the same r=206,477 target